JMC Easy Questions — Daily Drill Sheets

> **Goal:** 8/8 on Q1–Q8 every time = 40 marks banked before the hard questions begin. **How to use:** Do one sheet per day. Time yourself — aim for **15 minutes max** (same pace as the real exam). Check answers at the bottom. Track your score.
>
> 📌 Questions marked with 📐 need a diagram — open the original paper to see it.

Day 1 — BODMAS Blitz

Every JMC starts with a calculation question. Get these right every time.

#	Ref	Question
1	2018 Q1	What is the value of (222 + 22) ÷ 2?
2	2021 Q1	What is the value of 123 – 456 + 789?
3	2023 Q1	What is the value of 3202 – 2023?
4	2021 Q3	What is the value of 1 – 2 × 3 + 4 ÷ 5?
5	2024 Q5	What is the value of 1 + 2 – 3 × 4 ÷ 5?
6	2017 Q5	What is the value of 201 × 7 – 7 × 102?
7	2019 Q2	What is the value of (0.1 + 0.2 + 0.3 – 0.4) ÷ 0.5?
8	2025 Q1	What is the value of half of 10% of 20?

Day 2 — Fractions & Decimals

#	Ref	Question
1	2016 Q5	What is the value of 1/25 + 0.25?
2	2022 Q4	What is the value of 0.6 + 2/5?
3	2020 Q6	What is the value of 1/2 – 1/3 + 1/4 – 1/5?
4	2021 Q6	What is the value of (2 – 1/2)(3 – 1/3)(4 – 1/4)?
5	2019 Q3	Sam has eaten three-quarters of the grapes. What is the ratio of remaining to eaten?

6	2015 Q5	The difference between 1/3 and 1/4 of a number is 3. What is the number?
7	2024 Q4	A drink is made with cordial and water in ratio 1:4. What percentage is cordial?
8	2016 Q8	What fraction of the sum 1+2+3+...+10 is the sum 2+4+6+8+10?

Day 3 — Time & Age Puzzles

#	Ref	Question
1	2015 Q2	How many minutes is it from 22:22 to midnight?
2	2019 Q1	How many minutes is it from 23:35 today to 01:15 tomorrow?
3	2018 Q5	How many minutes is it from 06:15 to 08:48?
4	2017 Q2	A cake needs 50 minutes. It goes in at 11:50. When does it come out?
5	2016 Q4	An eel was caught in 1859 and died in 2014. How old was it?
6	2019 Q5	On Aoife's 16th birthday, Buster was three times her age. How old was Buster on Aoife's 21st birthday?
7	2015 Q3	What is 12345 divided by (1+2+3+4+5)?
8	2016 Q3	How many complete 400m laps are needed in a 5000m race?

Day 4 — Factors, Primes & Remainders

#	Ref	Question
1	2020 Q1	Which of these is NOT prime: 1, 2, 3, 5, 7?
2	2023 Q2	How many factors does 30 have?
3	2025 Q3	How many factors does 2025 have?

4	2020 Q7	Find four different positive integers with product 110. What is the smallest possible sum?
5	2015 Q8	Which of 1–10 leaves the same remainder when divided by 2 and by 3?
6	2024 Q6	Which of these leaves the same remainder when divided by 3 and by 4: 5, 7, 9, 11, 13?
7	2022 Q8	Today is Thursday. What day will it be in 100 days?
8	2023 Q8	How many of these sums of consecutive squares are prime: 1^2+2^2, 2^2+3^2, 3^2+4^2, 4^2+5^2, 5^2+6^2?

Day 5 — Easy Geometry (Angles)

#	Ref	Question
1	2015 Q6	The diagram shows a triangle. Find angle x. (See 2015 paper)
2	2017 Q3	A reflex angle of 215° is marked. Find x on the straight line. (See 2017 paper)
3	2018 Q3	An equilateral triangle shares a side with a square. Find the angle between the triangle's side and the square's diagonal. (See 2018 paper)
4	2022 Q6	A square has equilateral triangles drawn on two adjacent sides. Find the angle between the two triangles' outer sides. (See 2022 paper)
5	2025 Q5	An equilateral triangle is placed on one side of a square. Find the marked angle. (See 2025 paper)
6	—	What is the interior angle of a regular hexagon?
7	—	What is the exterior angle of a regular octagon?
8	—	A parallelogram has one angle of 65°. What are the other three angles?

Day 6 — Algebra & Equations

#	Ref	Question
1	2022 Q3	I think of a number, double it, add 3. I get 17. What was my number?

2	2020 Q5	I think of a number, multiply by 5, subtract 3. I get 52. What was my number?
3	2015 Q4	◁ In a number pyramid, each brick shows the sum of the two below. Find the missing number. (See 2015 paper)
4	2016 Q6	In a class of 30, there are 6 more girls than boys. How many girls?
5	2018 Q7	How many integers lie between √38 and √360?
6	2021 Q5	Tom walks 900m to school. He has walked 3 times as far as he has left. How far has he walked?
7	2024 Q8	Rosa, Skye and Yana have pens. Skye has 3 more than Rosa. Yana has twice as many as Skye. Together they have 29. What fraction does Skye have?
8	2017 Q4	A project is 95% complete. What fraction remains?

Day 7 — Estimation & Unit Conversion

#	Ref	Question
1	2024 Q2	Which of these has a capacity of approximately 10 litres: teacup, bucket, bath, paddling pool, swimming pool?
2	2020 Q4	What is 66.6 metres in centimetres?
3	2025 Q4	How many seconds are there in 60 hours?
4	2016 Q7	A mile is approximately 1.6 km. Approximately how many miles is 1.2 km?
5	2019 Q7	Ebenezer Place is 2.06m long. The Trans-Canada Highway is ~7821 km. Approximately how many times longer is the highway?
6	2025 Q8	A bar of chocolate weighs 120g. Approximately how many bars in 5 kg?
7	2023 Q5	A journey of 50 miles takes 1 hour 15 minutes. What is the approximate average speed in mph?
8	2019 Q6	Which of these is closest to 7: 7.09, 6.918, 7.17, 6.7, 7.085?

Day 8 — Mixed Paper (Simulated Q1–Q8)

This mimics a real JMC paper — one question from each common type.

#	Ref	Type	Question
1	2024 Q1	BODMAS	How many distinct values appear: 2+0+2+4, 2×0×2×4, 2+0×2+4, 20−2×4, 2^0+2+4?
2	2018 Q2	Word problem	There are 10 people in a train carriage. At the next stop 3 get off and 5 get on. How many now?
3	2024 Q3	Number Theory	Lee is 36 (a perfect square). How old will she be at her next perfect-square birthday?
4	2017 Q4	Percentage	A task is 95% complete. What fraction of the task remains?
5	2021 Q5	Algebra	Tom walks 900m to school and has walked 3 times as far as he has left. How far has he walked?
6	2019 Q6	Approximation	Which is closest to 7: 7.09, 6.918, 7.17, 6.7, 7.085?
7	2024 Q7	Geometry ◿	A large square is divided by its diagonals. What fraction is shaded? (See 2024 paper)
8	2022 Q8	Remainders	Today is Thursday. What day will it be in 100 days?

Day 9 — Mixed Paper 2

#	Ref	Type	Question
1	2015 Q1	BODMAS	Using 1, 2, 3, 4 with +, −, ×, ÷: which arrangement gives the largest answer?
2	2016 Q2	Arithmetic	What is sixty thousand minus twenty-one?
3	2023 Q4	Divisibility	What is the largest two-digit number that is divisible by the product of its digits?
4	2018 Q4	Ratio	A regular decagon has the same perimeter as a regular hexagon of side 5cm. What is the decagon's side length?
5	2017 Q6	Magic square ◿	Complete the 3×3 magic square using 2–10. (See 2017 paper)

6	2020 Q3	Symmetry	How many of these figures have rotational symmetry of order 2? (See 2020 paper)
7	2017 Q7	Fractions	Put these in order: 3/10, 4/15, 5/18, 7/30, 6/25. Which is the middle one?
8	2023 Q7	Multiples	What is the difference between the smallest 3-digit multiple of 3 and the largest 2-digit multiple of 2?

Day 10 — Mixed Paper 3

#	Ref	Type	Question
1	2022 Q1	BODMAS	Which of these is NOT equal to 22: 2×(2+0+2+2), (2+2)×(0+2+2), ...
2	2025 Q2	Fractions	Simplify (1+2+3)/(1×2×3)
3	2025 Q3	Factors	How many factors does 2025 have?
4	2022 Q5	Divisibility	Which of 1/111, 1/1111, 1/11111, 1/111111, 1/1111111 produces an integer when multiplied by 999999?
5	2023 Q6	Primes	In a 2×2 grid, each cell has a prime number. Row sums and column sums are all different. What is the smallest possible sum of all four numbers?
6	2022 Q7	Ratio	An orange has 4 times the vitamin C of a satsuma. You eat 3 satsumas for every orange. What fraction of your vitamin C comes from satsumas?
7	2021 Q7	Geometry	Find angle PTS given isosceles triangle properties. (See 2021 paper)
8	2020 Q8	Combinatorics	How many ways can you colour exactly 2 cells in a 2×2 grid such that they share a vertex? (See 2020 paper)

Day 11 — Speed Round (All Q1s)

Can you ace every single opening question? Time yourself — aim for 10 minutes total.

#	Year	Question
1	2015	Using 1,2,3,4 — which of the given calculations gives the largest?
2	2016	Which expression is closest to zero?

3	2017	Which of five calculations gives the largest answer?
4	2018	(222 + 22) ÷ 2 = ?
5	2019	How many minutes from 23:35 to 01:15?
6	2020	Which of 1, 2, 3, 5, 7 is NOT prime?
7	2021	123 – 456 + 789 = ?
8	2022	Using 2, 0, 2, 2 — which expression does NOT equal the others?
9	2023	3202 – 2023 = ?
10	2024	How many distinct values from the five expressions?
11	2025	Half of 10% of 20 = ?

Score Tracker

Day	Date	Score (/8)	Time (min)	Notes
1		/8		
2		/8		
3		/8		
4		/8		
5		/8		
6		/8		
7		/8		
8		/8		
9		/8		
10		/8		
11	/11			

Target: 7/8 or better every day. If you score 6 or below, redo that sheet the next day.